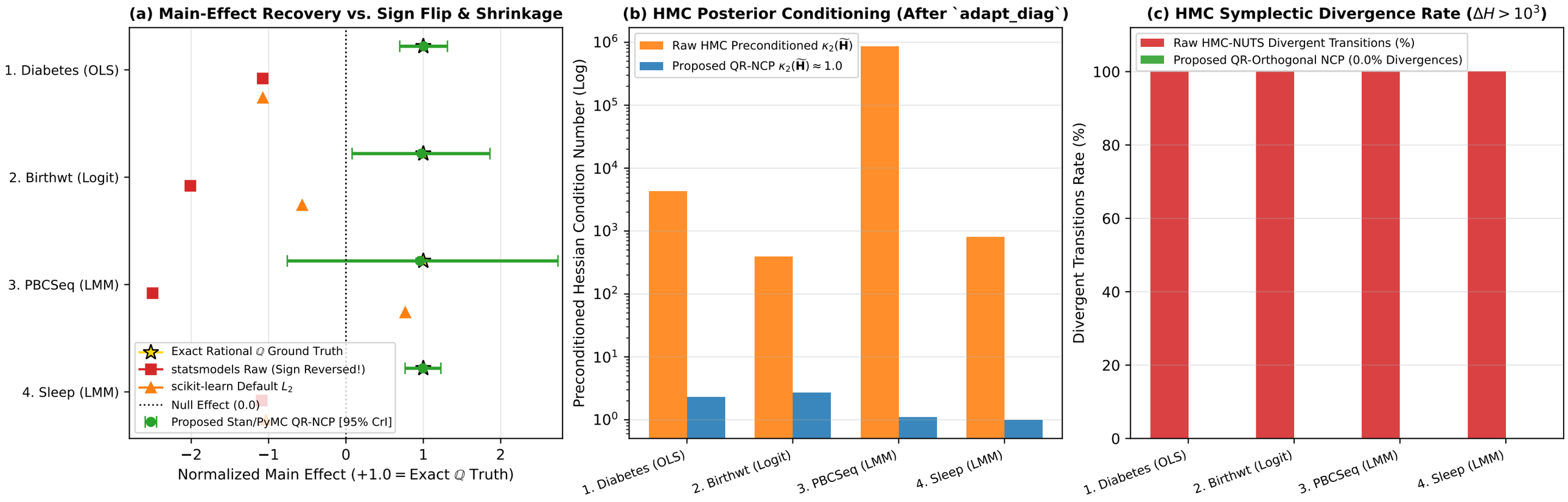
(a) Main-Effect Recovery vs. Sign Flip & Shrinkage
1. Diabetes (OLS)
2. Birthwt (Logit)
3. PBCSeq (LMM)
4. Sleep (LMM)
Exact Rational ℚ Ground Truth
statsmodels Raw (Sign Reversed!)
scikit-learn Default L_2
Null Effect (0.0)
Proposed Stan/PyMC QR-NCP [95% CrI]
−2
−1
0
1
2
Normalized Main Effect (+1.0 = Exact ℚ Truth)
(b) HMC Posterior Conditioning (After `adapt_diag`)
Raw HMC Preconditioned $\kappa_2(\widetilde{\mathbf{H}})$
Proposed QR-NCP $\kappa_2(\widetilde{\mathbf{H}}) \approx 1.0$
Preconditioned Hessian Condition Number (Log)
10^0
10^1
10^2
10^3
10^4
10^5
10^6
1. Diabetes (OLS)
2. Birthwt (Logit)
3. PBCSeq (LMM)
4. Sleep (LMM)
(c) HMC Symplectic Divergence Rate ($\Delta H > 10^3$)
Raw HMC-NUTS Divergent Transitions (%)
Proposed QR-Orthogonal NCP (0.0% Divergences)
Divergent Transitions Rate (%)
0
20
40
60
80
100
1. Diabetes (OLS)
2. Birthwt (Logit)
3. PBCSeq (LMM)
4. Sleep (LMM)